Essentials of Survey Design: Weightings

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Loading Data

```
library(haven)
LabourForceSurvey <- read_dta("C:/Users/c82254mm/Dropbox (The University of Manchester)/SOST20022/SOST2
```

Pick 2 continuous varibales:

```
class(LabourForceSurvey$AGE)
```

```
## [1] "haven_labelled" "vctrs_vctr"     "double"
```

```
LabourForceSurvey$AGE[LabourForceSurvey$AGE==0]<-NA
table(LabourForceSurvey$AGE)
```

```
##
##    1    2    3    4    5    6    7    8    9   10   11   12   13   14   15   16
## 1153 1256 1330 1377 1299 1292 1292 1275 1202 1197 1197 1193 1130 1108 1127 1111
##   17   18   19   20   21   22   23   24   25   26   27   28   29   30   31   32
## 1102 1011  950  955  935  967  942  929  963  994  977 1112 1030 1155 1146 1102
##   33   34   35   36   37   38   39   40   41   42   43   44   45   46   47   48
## 1197 1159 1277 1193 1127 1149 1196 1141 1207 1227 1251 1273 1368 1359 1339 1294
##   49   50   51   52   53   54   55   56   57   58   59   60   61   62   63   64
## 1379 1370 1419 1337 1333 1223 1243 1240 1202 1215 1175 1158 1109 1137 1154 1140
##   65   66   67   68   69   70   71   72   73   74   75   76   77   78   79   80
## 1083 1220 1165 1349 1243 1075 1115  949  965  760  532  357  384  338  247  225
##   81   82   83   84   85   86   87   88   89   90   91   92   93   94   95   96
##  188  171  147  143  131  108  110   70   62   49   45   28   33   20   10    7
##   97   98   99
##    3    2   10
```

```
class(LabourForceSurvey$BACTHR)
```

```
## [1] "haven_labelled" "vctrs_vctr"     "double"
```

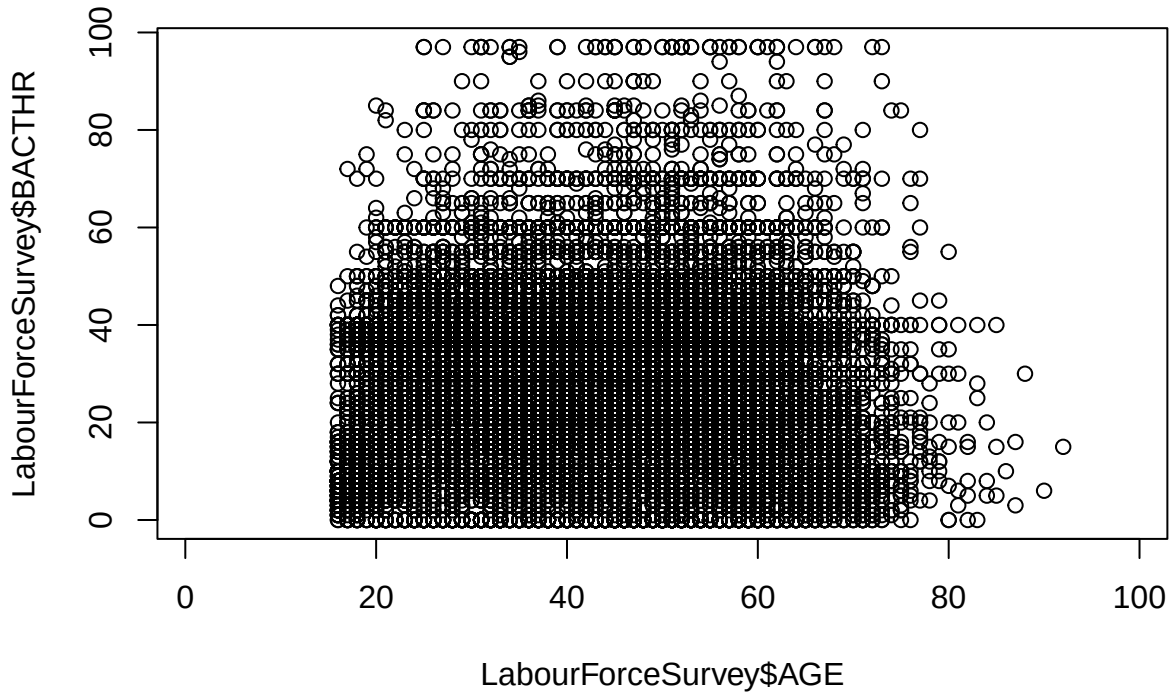
```
LabourForceSurvey$BACTHR[LabourForceSurvey$BACTHR<0]<-NA
table(LabourForceSurvey$BACTHR)
```

```
##
##    0    1    2    3    4    5    6    7    8    9   10   11   12   13   14   15
## 1021   45  125  120  229  190  294  168  562  123  563  101  560   93  288  647
##   16   17   18   19   20   21   22   23   24   25   26   27   28   29   30   31
## 1192  129  511  111 1282  396  731  202  975  765  253  202  703  137 2219  113
##   32   33   34   35   36   37   38   39   40   41   42   43   44   45   46   47
## 1072  176  277 2476 1067 2361 4430  848 6371   98  626  100  200 1185  107   74
##   48   49   50   51   52   53   54   55   56   57   58   59   60   61   62   63
```

```
##  567   35 1093   21   54   25   44  267   58   14   27    5  610    5   15   15
##   64   65   66   67   68   69   70   72   74   75   76   77   78   80   82   83
##    7   78   12    4   13    4  172   27    3   31    4    6    5   59    3    1
##   84   85   86   87   90   94   95   96   97
##   45   10    2    1   18    2    2    1   58
```

Calculate correlation:

```
plot(LabourForceSurvey$AGE, LabourForceSurvey$BACTHR)
```




```
cor(LabourForceSurvey$AGE, LabourForceSurvey$BACTHR)
```

```
## [1] NA
```

If your data contain missing values, use the following R code to handle missing values by case-wise deletion.

```
cor(LabourForceSurvey$AGE, LabourForceSurvey$BACTHR, use='complete.obs')
```

```
## [1] -0.003374412
```

Alternative way:

```
cor.test(LabourForceSurvey$AGE, LabourForceSurvey$BACTHR, use='complete.obs')
```

```
##
##  Pearson's product-moment correlation
##
## data:  LabourForceSurvey$AGE and LabourForceSurvey$BACTHR
## t = -0.67437, df = 39939, p-value = 0.5001
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.01318109  0.00643291
## sample estimates:
##          cor
## -0.003374412
```

```r
table1  <- table(LabourForceSurvey$AGE, LabourForceSurvey$BACTHR)
chisq.test(table1)
```

```
## Warning in chisq.test(table1): Chi-squared approximation may be incorrect
```

```
##
##  Pearson's Chi-squared test
##
## data:  table1
## X-squared = NaN, df = 8624, p-value = NA
```

Pick a categorical variable:

```r
class(LabourForceSurvey$RELIG11)
```

```
## [1] "haven_labelled" "vctrs_vctr"     "double"
```

```r
table(LabourForceSurvey$RELIG11)
```

```
##
##    -8     1     2     3     4     5     6     7     8
##  3715 30558 48609   396  1235   369  4027   557  1321
```

```r
LabourForceSurvey$RELIG11[LabourForceSurvey$RELIG11<0]<-NA  #Simple Recoding

LabourForceSurvey$New_Religion <- NA # Create new variable
LabourForceSurvey$New_Religion[LabourForceSurvey$RELIG11==1] <- "No Religion"
LabourForceSurvey$New_Religion[LabourForceSurvey$RELIG11>1] <- "Religious"
table(LabourForceSurvey$New_Religion)
```

```
##
## No Religion   Religious
##       30558       56514
```

TESTING THE DIFFERENCE BETWEEN MEANS - t.test()

H_0: Religious and Non-Religious people have the same actual hours in their main job H_1: Religious and Non-Religious people do not have the same actual hours in main job

```r
t.test(LabourForceSurvey$BACTHR[LabourForceSurvey$New_Religion=="No Religion"],
       LabourForceSurvey$BACTHR[LabourForceSurvey$New_Religion=="Religious"])
```

```
##
##  Welch Two Sample t-test
##
## data:  LabourForceSurvey$BACTHR[LabourForceSurvey$New_Religion == "No Religion"] and LabourForceSurv
## t = 5.2822, df = 31517, p-value = 1.285e-07
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  0.4753003 1.0361420
## sample estimates:
## mean of x mean of y
##  32.81525  32.05953
```

Weighted t-test:

```r
# install.packages('weights')
library(weights)
```

```
## Loading required package: Hmisc
```

```
##
## Attaching package: 'Hmisc'
```

```
## The following objects are masked from 'package:base':
##
##     format.pval, units
```

```
wtd.t.test(x = LabourForceSurvey$BACTHR[LabourForceSurvey$New_Religion=="No Religion"],
           y = LabourForceSurvey$BACTHR[LabourForceSurvey$New_Religion=="Religious"],
           weight=LabourForceSurvey$PWT18[LabourForceSurvey$New_Religion=="No Religion"],
           weighty=LabourForceSurvey$PWT18[LabourForceSurvey$New_Religion=="Religious"])
```

```
## Warning in wtd.t.test(x =
## LabourForceSurvey$BACTHR[LabourForceSurvey$New_Religion == : Treating data for
## x and y separately because they are of different lengths
```

```
## $test
## [1] "Two Sample Weighted T-Test (Welch)"
##
## $coefficients
##      t.value           df      p.value
## 4.853799e+00 3.297018e+04 1.216718e-06
##
## $additional
## Difference     Mean.x     Mean.y   Std. Err
##  0.6821308 33.0365703 32.3544395  0.1405355
```

Pick another categorical variable:

```
class(LabourForceSurvey$ETHEWEUL)
```

```
## [1] "haven_labelled" "vctrs_vctr"     "double"
```

```
table(LabourForceSurvey$ETHEWEUL)
```

```
##
##    -9    -8     1     2     3     4     5     6     7     8     9    10    11
## 10641    86 64580   459  4264   489   196   338   377  2048  1593   591   385
##    12    13    14    15    16
##  1033  1524   781   170  1232
```

```
LabourForceSurvey$ETHEWEUL[LabourForceSurvey$ETHEWEUL<0]<-NA  # simple recoding

LabourForceSurvey$New_Ethnicity <- NA #create new variable
LabourForceSurvey$New_Ethnicity[LabourForceSurvey$ETHEWEUL>=1 & LabourForceSurvey$ETHEWEUL<=3] <- "Whit
LabourForceSurvey$New_Ethnicity[LabourForceSurvey$ETHEWEUL>4] <- "Non-White"

table(LabourForceSurvey$New_Ethnicity)
```

```
##
## Non-White     White
##     10268     69303
```

TESTING the Independence BETWEEN Two Categories - `chisq.test()`

H_0: Religion and Ethnicity are indepdendent. H_1: Religion and Ethnicity are NOT indepdendent.

```
chisq.test(LabourForceSurvey$New_Ethnicity,LabourForceSurvey$New_Religion)
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  LabourForceSurvey$New_Ethnicity and LabourForceSurvey$New_Religion
## X-squared = 2752.4, df = 1, p-value < 2.2e-16
```

Weighted Chi-squared test:

```
wtd.chi.sq(LabourForceSurvey$New_Ethnicity, LabourForceSurvey$New_Religion,
           weight=LabourForceSurvey$PWT18, na.rm=TRUE)
```

```
##    Chisq       df  p.value
## 2809.149    1.000    0.000
```